



HELIONEST

SOLAR COOL UMBRELLA HELIONEST



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EXECUTIVE SUMMARY HELIONEST

In the Philippines, many students and commuters feel exhausted and even sick from the heat between 10am and 3pm. A regular umbrella blocks the sun but makes you feel hotter because there's no airflow. Our product, the solar cool umbrella, adds a small fan powered by the sun's light using solar panels. It still blocks 99% of UV rays and even lets you charge your phone through a USB port. No batteries needed, just sunlight. We're starting in Pampanga, Manila, and Cebu, where over 5 million people face this problem every day. We expect to sell 30,000 units in year 1 at 1,299 each, bringing in about 39 million. Our startup costs are 925,000, covering design, first inventory, marketing, and legal fees. We plan to sell online (Shopee, Lazada, TikTok Shop, etc.) and through physical stores. With a simple and real solution to beating the heat, the solar cool umbrella can keep Filipinos safer and more comfortable while giving investors a clear path to growth.

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COMPANY PROFILE

HELIONEST

A sanctuary of smart, solar-powered comfort.

VISION

At Helionest Inc., we innovate sustainable solar-powered solutions that enhance comfort, convenience, and eco-friendly living. Through cutting-edge technology and smart design, we empower individuals and communities to enjoy seamless, energy-efficient lifestyles.

MISSION

Helionest Inc. aims to lead solar innovation worldwide, creating a future where sustainability and comfort go hand in hand. We envision smart energy solutions that transform everyday spaces into efficient, eco-friendly havens powered by the sun.

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PROBLEM AND SOLUTION

SOLARCOOL UMBRELLA

THE PROBLEM

In the Philippines, many people especially the Students, Workers, and Commuters face a serious discomfort and health risks while walking, or working outdoors between 10am to 3pm. as ordinary umbrellas they only provide UV protection but fail to keep them cool, leading to tiredness, dizziness, or even sickness.

THE ROOT CAUSE

The cause of the problem is that the traditional umbrellas are only designed to block sunlight, not to reduce heat or provide airflow, so many people can't stay cool in extreme outdoor temperatures.

THE SOLUTION

The Solar Cool Umbrella solves the problem by combining sun protection and solar powered cooling. It's built for hot days and helps people feel more comfortable while they walk, or work outdoors.

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MARKET ANALYSIS

SOLARCOOL UMBRELLA

1. INDUSTRY OVERVIEW

A. Industry Size and Growth Rate

Traditional umbrellas are no longer enough to protect students and commuters from intense sun exposure, which is why more people are looking for better options like the SolarCool Umbrella. It cools you down with solar power and helps protect against harmful sunlight.

Globally, The price of the standard umbrella ranges from \$15 to 20\$ depends on the brand and the design. According to MarketWatch(2017), people spend within this range for a quality umbrella that only offers basic protection. This shows that the consumers are more willing or interested in new innovative products like the SolarCool Umbrella with added features of cooling and solar.

Here in the Philippines, regular umbrellas are generally 100-300 pesos , depending on quality and brands that can be bought on the streets or online shops (ToplistPH, 2023). This affordable price is accessible to many but it also highlights growing interest in something new or upgraded alternatives. As people spend more time on outdoor activities in hot weather there's a high chance of high demand for SolarCool Umbrella.

B. Life Cycle Stage

The tech-industry is growing through the integration of smart features into ordinary things like umbrellas with built-in cooling. As climate change causes more intense heat, people desire practical, technologically based strategies to be cool. This demonstrates how technology is utilized to better daily life.

C. Trends and Developments

- **Eco-Friendly and Sustainable Product** - Mostly Students and Commuters choose the solar-powered umbrella with multiple features without using electricity.
- **Portable and multi-functional tools** - Consumers preferred more functionalities that helped their comforts.



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MARKET ANALYSIS

SOLARCOOL UMBRELLA

1. INDUSTRY OVER VIEW

D. Key Success Factors

To succeed in the outdoor tech and smart accessory industry with a product like the Solarcool Umbrella, a business must prioritize fast innovation. The market needs aspects like solar efficiency, battery storage, and user comfort to keep getting better, therefore agile product development is a must. Low costs can help you reach more people, but the most important thing is to offer good value. This means giving customers many perks, including solar-powered cooling, UV protection, and charging their devices, that make the higher price worth it. Network effects are less relevant unless the device becomes a linked platform. But good customer service, being eco-friendly, and smart design can help you build a strong brand that will last.

2. TARGET MARKET ANALYSIS

A. Market Segmentation

- Demographics
 - Age: 15-45 years old
 - Occupation: Students, commuters, street vendors, travelers.
 - Income Level: Low to middle-income earners
- Psychographics
 - Health and comfort
 - Environmental friendly
 - Smart solutions
- Behavior
 - Prone purchasing for summer season.
 - Social media ads promotion
- Geographic location
 - Pampanga, Manila and Cebu



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MARKET ANALYSIS

SOLARCOOL UMBRELLA

2. TARGET MARKET ANALYSIS

B. Target Market Definition

Our primary Customer is Filipino students, commuting daily to school under extreme heat. They walk or ride public transportation, and mostly wait at some hot areas waiting for jeepneys.

C. Customer Needs and Pain Points

- **Overexposure to Sunlight** - Students and commuters get tired and dizzy after long exposure to sunlight.
- **Traditional umbrella trap heat** - regular umbrellas block sunlight but it traps the warm air making it more hotter.
- **Bulky Cooling Gadgets** - mini fans are bulky and aren't convenient during commutes.
- **Risk of Heat Stroke** - sun exposures of students can cause dehydration especially for students walking to school.

D. Customer Buying Behavior

The product is primarily bought online during summer or back-to-school season, driven by real-user reviews, influencer endorsements, and its affordable price. Social media platforms like TikTok and Facebook, along with online shops like Shopee and Lazada, play a key role in promoting its cooling and sun protection benefits.

3. MARKET SIZE AND GROWTH

A. Total Addressable Market (TAM)

The Total Addressable Market (TAM) for the Solarcool Umbrella includes the worldwide demand for solar-powered outdoor umbrellas that can do more than one thing. This is part of the quickly increasing solar umbrella market. This market is growing as a result of growing consumer interest in tech-integrated, sustainable products.

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MARKET ANALYSIS

SOLARCOOL UMBRELLA

3. MARKET SIZE AND GROWTH

A. Total Addressable Market (TAM)

Demand for new technologies like the Solorcool Umbrella is growing as a result of growing awareness of UV protection, the need for mobile charging, and climate-conscious living, particularly in areas with loads of direct sunlight. The TAM offers a significant and scalable opportunity for global adoption as outdoor recreation and smart living trends continue to expand globally.

B. Serviceable Available Market (SAM)

The SolarCool Umbrella is more focused on the Philippine market, hence there are over 30 million students and an estimated 25 million commuters who are frequently exposed to sunlight (PSA, 2023). If at least 10-15% of this group is targeted mostly around 5-8 million individuals, the product has a high opportunity to be marketable. Through the right promotion and accessibility the product would become mainstream personal cooling essential.

C. Serviceable Obtainable Market (SOM)

Within the first 1-3 years, the product is effective in marketing, affordable and successful. The business has 50,000 to 100,000 customers in areas of Pampanga, Manila and Cebu. This is especially during summer seasons or school openings.

D. Market Growth Rate

The market for solar umbrellas, which includes unique technologies like the Solorcool Umbrella, is rising rapidly due to increased customer need for environmentally conscious, trustworthy, usable outdoor solutions. According to Business Research Insights, the market for sunny umbrellas, which include solar-powered models, will expand at a Compound Annual Growth Rate (CAGR) of 5.2%, from USD 3.64 billion in 2024 to USD 5.75 billion by 2033. Growing consumer interest in solar-powered charging, UV protection, and features like cooling fans is driving this expansion.

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MARKET ANALYSIS

SOLAR COOL UMBRELLA

4. COMPETITIVE ANALYSIS

A. Direct vs. Indirect Competitors

Direct competitors include regular umbrellas, which are inexpensive but lack cooling features, and handheld mini fans that provide portable airflow. Indirect competitors include hats and cooling towels, which offer passive heat relief through shade or moisture-based cooling.

B. Strengths and Weaknesses (SWOT)

Strengths

1. Eco-Friendly: Solar-powered, reducing reliance on batteries or electricity.
2. Innovative Cooling: Unique cooling functionality provides relief from heat.
3. Portable: Umbrella design makes it easy to carry and use outdoors.
4. Sustainable: Renewable energy source reduces carbon footprint.
5. Cost-Effective: Solar power can save money on energy costs.
6. Multi-Functional: Provides shade and cooling, making it a practical product.
7. Attractive Design: Modern and stylish, appealing to environmentally conscious consumers.

Weaknesses

1. Dependence on Sunlight: Performance affected by weather conditions (cloudy days).
2. High Initial Cost: Advanced technology and materials may increase production costs.
3. Limited Cooling Capacity: May not be effective in extremely hot temperatures.
4. Battery Storage: May require battery storage for nighttime or cloudy days.
5. Durability Concerns: Umbrella frame and solar panels may be prone to damage.
6. Maintenance Requirements: Regular cleaning of solar panels necessary for optimal performance.
7. Weight: Solar panels and battery storage may add weight.



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MARKET ANALYSIS

SOLAR COOL UMBRELLA

4. COMPETITIVE ANALYSIS

B. Opportunities and Threats (SWOT)

Opportunities

1. High Demand: People want more eco-friendly products.
2. Outdoor Use: Great for events, beaches, and travel.
3. Eco Partnerships: Team up with green-minded brands.
4. New Markets: Sell in hot, sunny regions.
5. Modern Marketing: Use social media and influencers.
6. More Products: Add items like solar chargers.
7. Govt Perks: Explore rebates or incentives.

Threats

1. Big Brands Lead: Famous companies already sell similar products.
2. Fast-Changing Tech: New inventions could replace yours.
3. Strange Weather: Unpredictable weather might make it harder to use.
4. Low Buying Power: People might skip extras when money's tight.
5. Rule Changes: New laws could affect solar-powered items.
6. Fake Products: Cheap copies can harm your brand's reputation.
7. Delivery Trouble: Issues with shipping or supplies can cause delays.

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MARKET ANALYSIS

SOLARCOOL UMBRELLA

4. COMPETITIVE ANALYSIS

C. Barriers to Entry

In order to safeguard the company and gain a competitive advantage in the market, the Solorcool Umbrella benefits from a number of important barriers to entry. First, the integration of efficient solar panels, battery systems, and cooling mechanisms, among other high technology costs, creates a financial barrier for new entrants trying to replicate the product's quality and functionality. Second, intellectual property for exclusive fan designs, energy management components, and solar-powered cooling systems are examples of intellectual property that can legally protect innovation and stop direct copying. Lastly, brand loyalty is important. By delivering a trustworthy, high-performing product and developing a strong brand identity focused on comfort, sustainability, and smart outdoor living, Solorcool may develop a loyal customer base that is less likely to move to competing businesses. When combined, these barriers contribute to creating a strong position in a specific developing market.

5. REGULATORY AND ENVIRONMENTAL FACTORS

A. Technological Environment

Implementation of new technologies such as the 5G network, AI, and IoT could help SolarCool Umbrella enhance its functionalities, creating real-time weather notifications and intelligent cooling controls. With regard to the user experience, future models might integrate with applications. On the flip side, competitors could introduce either cheaper solar technology or faster innovations, thereby disrupting the entire market. Therefore, the company must keep tracking the latest technological trends.



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MARKET ANALYSIS

SOLARCOOL UMBRELLA

6. DISTRIBUTION CHANNELS AND SUPPLY CHAIN

A. Channel Types

- **Online store** - The Solar Cool Umbrella will be sold in online stores like Shopee and Lazada.
- **Direct B2B sales** - Like Schools, Churches, and Sidewalks can order directly through B2B sales channels.
- **Partner resellers** - local resellers will help to reach more customers in the community.

B. Logistics and Supply Chain

- **For physical products:**
 - **Manufacturers** - Manufacturers will produce the umbrella parts and assemble it to the final product.
 - **Distributors** - This will handle shipping and delivery to stores and direct buyers.
 - **Retailers** - Will offer the product in physical stores and online shops for easy access.

- **For digital products:**

The Solorcool Umbrella's digital features, like the website or app, need reliable hosting to guarantee uptime, user support systems to address queries or problems, and routine maintenance to keep the platform current and functional.

7. PRICING AND POSITIONING STRATEGY

A. Pricing Models

The SolarCool Umbrella will follow a one-time purchase model, starting priced at 850 - 1299 PHP per unit. This price reflects on its functionality like offering sun protection with active cooling and eco-friendly solar power, the discounts may be offered during summer season.



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MARKET ANALYSIS

SOLARCOOL UMBRELLA

7. PRICING AND POSITIONING STRATEGY

B. Differentiation and Unique Selling Proposition (USP)

Unlike the regular umbrella the SolarCool Umbrella offers cooling and sun protection in one product. It's also eco-friendly and rechargeable by the sun. It is designed for daily use in extreme hot weather.

8. DEMAND FORECASTING

- **Predict customer adoption using:**
 - **Surveys** - Show that most students and commuters say they struggle with the heat when outdoors.
 - **Pre-orders** - During School openings and Summer months, when people are most exposed to heat and actively looking for ways to stay cool while in outdoor activity.
 - **Comparable product adoption curves** - Similar products like neck fans and UV umbrellas become popular during hot months and school openings, so we expect that the solar cool umbrella is to follow the same trend.

YEAR	ESTIMATED UNIT SOLD	REVENUE (PHP)	KEY DEMANDS
Year 1	30,000 units	P29M - P38M	Launch campaigns, Back to school.
Year 2	60,000 units	P50M - P70M	Strong online presence , Viral marketing.
Year 3	100,000 units	P99M - P120M	Brand loyalty, Expanded distribution nationwide.

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PRODUCT DESCRIPTION

SOLARCOOL UMBRELLA

1. PRODUCT OVERVIEW

- **Name:** SolarCool Umbrella
- **Type:** Solar Powered Umbrella with Built in fan.
- **Purpose:** SolarCool Umbrella provides a Solar Energy to power up the integrated fan. Ideal for Hot and Sunny Days, it keeps you comfortable while you are outdoors.

2. KEY FEATURES

- **Solar Powered Cooling Fan** - Built with high efficiency solar panels on the umbrella that converts sunlight to a portable energy that was getting stored in the batteries. The Cooling fan is located at the center pole of the umbrella. It's designed to circulate and keep the user cool.
- **UV Protection** - Blocks harmful UV rays from the sun, helping to protect your skin and reduce heat keeping you safer and cooler during long hours outdoors.
- **USB Charging Port** - The charging feature is providing an output of 5V/1A, that charges the devices such as mobile phones. This charging feature is connected to the battery where the converted solar energy is getting stored.

3. TECHNOLOGY STACK

Hardware Materials

- Solar Panels
- Voltage Regulator
- Cooling Fan
- Canopy Material
- Batteries
- Frame Material
- Usb Charging Port

- **Protection and Comfort** – The SolarCool Umbrella protects you from the harsh sunlight thanks to the fabric that has a UV Rays protection and the Built-In cooling fan of the said product.
- **Charging on the Go** – Thanks to the converted energy of the SolarCool Umbrella that was getting stored by a battery, the energy is getting preserved in the said battery. This battery will be the one supplying power so the mobile device can be charged if the battery has stored enough energy
- **Reliable Power-Source** – SolarCool Umbrella is relying on Solar Power Energy. The batteries are being charged even if you are on a walk outside. The user won't be worrying where to charge their device if they are outside.



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OPERATION PLAN

SOLARCOOL UMBRELLA

1. OBJECTIVES

Objective 1: Make people more comfortable outdoors with solar-powered cooling.

Objective 2: Promote eco-friendly solutions through innovative umbrella design.

Objective 3: Grow sales in hot, sunny regions and outdoor event markets.

2. TASKS/ACTIVITIES

Task/Activity	Description	Responsible Person/Team	Frequency
Design & Product Testing	Build and test umbrella features	Product Team	Monthly
Marketing & Promotion	Social media, influencers, ads	Marketing Team	Weekly
Sales & Distribution	Manage orders and deliveries	Sales Team	Daily
Customer Feedback & Support	Collect and respond to customer feedback	Customer Service Team	Ongoing



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OPERATION PLAN

SOLARCOOL UMBRELLA

3. TIMELINE

Activity	Start Date	End Date	Duration	Status
Product Finalization	AUG 2025	OCT 2025	3 MONTHS	In Progress
Marketing Launch	NOV 2025	DEC 2025	2 MONTHS	Up Coming
Initial Sales Rollout	JAN 2026	MAR 2026	3 MONTHS	Planned

4. RESOURCES

- **Human Resources:**
 - Designers, Engineers, Marketing Team, Sales Staff, Support Team
- **Physical Resources:**
 - Materials (fabric, solar panels, fans, rechargeable battery, USB charging port), Packaging, Storage facilities.
- **Technological Resources:**
 - Website, E-commerce platform, Social media tools.
- **Financial Resources:**
 - Budget for product development, marketing, operations.

5. ROLES/RESPONSIBILITIES

Role	Responsibility
Product Lead	Oversees design and performance testing
Marketing Manager	Runs ad campaigns and social media content



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OPERATION PLAN

SOLARCOOL UMBRELLA

5. ROLES/RESPONSIBILITIES

Role	Responsibility
Sales Coordinator	Handles orders and tracks deliveries
Customer Service Lead	Manages support and collects feedback
Prototype Developer	Assemble the product and improve it based on testing.

6. KEY PERFORMANCE INDICATORS (KPIs)

- KPI 1: Customer Satisfaction (via reviews & surveys)
- KPI 2: Sales Volume (units sold per month)
- KPI 3: Website Traffic (visits and engagement)
- KPI 4: Return Rate (percentage of returned products)
- KPI 5: Social Media Reach (followers, shares, likes)

7. CONTINGENCY PLAN

- Risk 1: Tech Failure (fan/USB)
 - Mitigation Strategy: Implement strict quality checks before shipping.
- Risk 2: Weather Irregularities
 - Mitigation Strategy: Promote multi-season use (e.g. light rain coverage)
- Risk 3: Supply Chain Delays
 - Mitigation Strategy: Secure backup suppliers and extra stock.
- Risk 4: Copycat Products
 - Mitigation Strategy: Apply for design patents and brand trademark.
- Risk 5: Economic Slowdown
 - Mitigation Strategy: Offer bundle deals and flexible pricing.


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FINANCIAL PLAN

SOLARCOOL UMBRELLA

1. START-UP COSTS

Item	Estimated Cost	Notes
Product Development	₱150,000	Design, prototyping with solar tech + UV fabric
Materials & Inventory	₱500,000	Initial stock for launch production
Marketing & Promotion	₱200,000	Launch campaigns, social media, back-to-school
E-commerce & Website	₱50,000	Online store setup, hosting, tools
Legal & Admin Fees	₱25,000	Permits, licenses, trademarks

2. SALES FORECAST

Period	Units Sold	Price per Unit	Total Revenue
Year 1	30,000	₱850–₱1,299	₱29M – ₱38M
Year 2	60,000	₱850–₱1,299	₱50M – ₱70M
Year 3	100,000	₱850–₱1,299	₱99M – ₱120M


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FINANCIAL PLAN

SOLARCOOL UMBRELLA

3. INCOME STATEMENT (PROJECTED)

Revenue :	₱29,000,000 – ₱38,000,000
Cost of Goods Sold (₱300/unit avg):	₱9,000,000
Operating Expenses:	₱4,000,000
Net Profit (Estimated):	₱16M – ₱25M

4. CASH FLOW STATEMENT

Period	Cash Inflows	Cash Outflows	Net Cash Flow
Q1	₱5M	₱3M	₱2M
Q2	₱9M	₱5M	₱4M
Q3	₱7M	₱4M	₱3M
Q4	₱8M	₱4.5M	₱3.5M


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FINANCIAL PLAN

SOLARCOOL UMBRELLA

5. BALANCE SHEET (PROJECTED)

ASSETS	
ASSETS	AMOUNT
CASH	₱6M
INVENTORY	₱2M
EQUIPMENT	₱1M

LIABILITIES & EQUITY	
ITEM	AMOUNT
Loans Payable	₱500,000
Owner's Equity	₱8.5M


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FINANCIAL PLAN

SOLARCOOL UMBRELLA

6. BREAK-EVEN ANALYSIS

Fixed Costs:	₱4,000,000
Selling Price per Unit:	₱1,100 (average)
Variable Cost per Unit:	₱300
Break-even Point (units):	$\text{₱4,000,000} \div (\text{₱1,100} - \text{₱300}) = \mathbf{5,000 \text{ units}}$

7. FUNDING REQUIREMENTS

Total Funds Needed:	₱925,000
Use of Funds:	
Product Development:	₱150,000
Inventory & Supplies:	₱500,000
Marketing & Setup:	₱250,000
Legal & Admin:	₱25,000

Preferred Source of Funding:

- Private investors, crowdfunding platforms, and government grants focused on eco-friendly innovation.


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CONCLUSION

SOLARCOOL UMBRELLA

The SolarCool umbrella is made to help people to stay cool and safe under the harsh sunlight. A lot of students and workers in the Philippines are walking and working outside during the hottest hours of the day. Umbrellas block sunlight, but they can't make you feel fresh and cooler. SolarCool Umbrella gives shade, cool air with a solar-powered fan, and can even charge your phone, all without the use of electricity. After studying the market, we found that many people are looking for new ways to stay cool outdoors, especially during summers and school openings. Market analysis showed us what people really need, what problems they face, and how our product can help. It gave us an idea on how to sell and how to promote it better. This kind of study lowers the risk and helps our business to grow. As more people looking for eco-friendly technology, our SolarCool Umbrella has a great potential to succeed and grow in the market.

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